CAS FACTORY · SUPERVISOR ROUTING

Model matrix recommendations

Date: 2026-07-09 · **Rev:** 2 (Composer as Grok Build model) · **Scope:** cas-supervisor model selection after first-class Grok harness (cas-8888)

Leaderboard snapshot primarily from **Artificial Analysis** (Intelligence Index v4.1, Coding Agent Index, model pages) as of 8–9 July 2026, plus live `grok models` availability on this host.

> **Rev 2 correction:** Composer 2.5 is *not* Cursor-only for factory use. On Grok Build (`grok` 0.2.93), available models are exactly `grok-4.5` (default) and `grok-composer-2.5-fast`. CAS can already spawn it via `cli=grok model=grok-composer-2.5-fast`.

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1. Executive recommendation

Primary decision

Run most factory engineering on the

Grok Build harness

with two models:

Composer 2.5 Fast

for light/fast work,

Grok 4.5

for standard and heavy. Keep

Claude Opus

as scarce frontier insurance. Stop treating

Claude Sonnet

as default heavy. Codex/Haiku become optional capacity backups, not the light default.

- **Light / flash:** `cli=grok model=grok-composer-2.5-fast`
- **Standard floor:** `cli=grok model=grok-4.5 effort=medium`
- **Heavy default:** `cli=grok model=grok-4.5 effort=high`
- **Frontier:** `cli=claude model=opus effort=high` (architecture, taste-max, 2× bounce)

Artificial Analysis places Grok 4.5 (high) at Intelligence Index **54** (near frontier), with Coding Agent Index performance **on par with GPT-5.5 in Codex** at roughly **half the $/task** and far fewer tokens. Composer 2.5 sits on the CAI cost/speed Pareto frontier (~$0.07–$0.44/task, ~6.7–9.3 min wall time in AA Cursor measurements) and is **already selectable in Grok Build** as `grok-composer-2.5-fast`.

2. Live Grok model availability (this host)

Verified 2026-07-09 via `grok models` on Grok Build `0.2.93`:

```
Default model: grok-4.5

Available models:
  * grok-4.5 (default)
  - grok-composer-2.5-fast
```

CAS factory already supports `cli=grok` and passes `-m <MODEL>` through `PtyConfig::grok`. No new harness is required to use Composer — only an explicit `model=` on `spawn_workers`.

> **Prior draft error (rev 1)**
>
> Rev 1 incorrectly stated Composer needed a Cursor harness. That conflated model origin (Cursor/Anysphere training lineage; AA measured in Cursor CLI) with runtime availability (Grok Build exposes it as a second model ID). Rev 2 corrects this.

3. Leaderboard evidence

3.1 Intelligence Index (quality ceiling)

Rank	Model	II score
1	Claude Fable 5 (max, Opus fallback)	60
2	Claude Opus 4.8 (max)	56
3	GPT-5.5 (xhigh)	55
4	**Grok 4.5 (high)**	**54**
~5	Claude Sonnet 5 (max)	53

Sources: Artificial Analysis · [C1][C2][C3].

3.2 Speed & list pricing

Model	Output speed	List $/M in · out	Notes
Grok 4.5 (high)	**91.3 tok/s**	$2 · $6	Cache hit $0.50 (−75%); TTFT ~16.5s (reasoning)
Composer 2.5 Fast	agent wall ~6.7 min/task (AA)	$3 · $15 (Fast token rates)	Fastest high-quality agent lane; Grok model id `grok-composer-2.5-fast`
Composer 2.5 Standard	agent wall ~9.3 min/task (AA)	$0.50 · $2.50	~$0.07/task on CAI (AA Cursor measure); not listed as separate Grok model id here
Claude Sonnet 5 (max)	~85 tok/s	$3 · $15	Can exceed Opus $/task due to verbosity
Claude Haiku 4.5	~97–127 tok/s	$1 · $5	Fastest Claude decode; weaker agent Pareto vs Composer
Claude Opus 4.8 (max)	~61–66 tok/s	$5 · $25	~1.4× slower decode than Grok 4.5

Sources: [C2][C4][C5][C6][C9].

3.3 Cost efficiency ($/task beats $/token)

Metric	Grok 4.5 (high)	Peers (AA)
$ / Intelligence Index task	**~$0.31**	GPT-5.5 xhigh ~$0.99; Opus max ~$1.78; Sonnet 5 max ~$2.29
Output tokens / II task	~14k (~60%+ less than Opus)	Sonnet 5 max uses ~40% more output tokens than Sonnet 4.6
Coding Agent Index (harness)	**76** in Grok Build	On par with GPT-5.5 xhigh in Codex; just under Fable 5 in Claude Code
$ / CAI task	**~$2.49–2.59**	GPT-5.5 Codex ~$5.07; Fable 5 Claude Code ~$11.80
Tokens / CAI task	**~1.9M**	GPT-5.5 Codex ~6.2M; Fable 5 ~7.2M (~3–4× more)
Composer 2.5 $/CAI task	**$0.07** standard · **$0.44** Fast (AA Cursor CLI measure) — best cost Pareto above CAI 60 [C9]	

Sources: [C1][C7][C8][C9].

4. What the old matrix got wrong

Current cas-supervisor rubric (`model-selection.md`):

Old tier	Old spawn	Board + availability verdict
light	`codex / gpt-5.5 / low`	**Replace with Composer on Grok.** `grok-composer-2.5-fast` is the Haiku-shaped lane with better agent quality/$ [C9][C13].
standard (floor)	`codex / gpt-5.5 / medium`	Grok 4.5 matches GPT-5.5 CAI class at roughly half $/task and ~⅓ the tokens. New floor is Grok medium [C1].
heavy	`claude / sonnet / high`	**Economically obsolete as default.** Sonnet 5 max can cost more per II task than Opus. Grok high is the rational heavy default [C1][C8].
frontier	`claude / opus / high`	Correct quality ceiling; wrong as frequent worker. CAI peers cost 2–5× Grok at similar coding-agent scores.

Form issue, not just slugs

Four capability tiers still work. Binding light/standard to Codex and heavy/frontier to Claude is outdated now that Grok Build is first-class and exposes both Composer and Grok 4.5 as models on one CLI.

5. Recommended matrix

5.1 Axes (include Speed)

Axis	Measure	Prefer
Intelligence	II + CAI	Opus/Fable only when +2–6 II points matter
Cost	**$/task**, not $/M tokens	Composer Fast, Grok 4.5, Codex subscription backup
Speed	TPS + agent wall time + tokens/task	Composer Fast first; then Grok 4.5
Taste	Supervisor judgment (unbenchmarked)	Claude for prose, skills, public APIs

5.2 Capability tiers (CAS-real spawns)

LIGHT

Flash / mechanical / small edits

```
cli=grok model=grok-composer-2.5-fast
```

Chores, docs, renames, config bumps, `depth=light`, pattern-mirror tests, "what Haiku would have done." Same Grok harness as the rest of the Grok fleet — only the model id changes. Fallback: `cli=codex model=gpt-5.5 effort=low` if Grok quota is exhausted.

STANDARD

New engineering floor

```
cli=grok model=grok-4.5 effort=medium
```

Clear AC, bounded blast radius. Fallback: `cli=codex model=gpt-5.5 effort=medium` for bulk parallel when Grok capacity is tight.

HEAVY

Cross-cutting / critical path

```
cli=grok model=grok-4.5 effort=high
```

Refactors, concurrency/lifecycle, P0/P1 path. Escalate to Sonnet only for taste or Claude-lifecycle thrash — and watch Sonnet $/task.

FRONTIER

Architecture & rescue

```
cli=claude model=opus effort=high
```

Architecture-defining units, high blast radius, twice-failed tasks, “would read the diff twice.” Not volume engineering.

5.3 Why this mapping (evidence → action)

- **Composer light:** available on Grok Build as the only non-default model [C13]; AA cost/speed Pareto leader for agents above CAI 60 [C9]; replaces Haiku-shaped work.
- **Grok standard/heavy:** II 54 near frontier; CAI ≈ GPT-5.5 Codex; ~$2.5/CAI task vs ~$5–12 peers; ~1.9M tokens vs 6–7M; 91 tok/s [C1][C2].
- **Opus frontier only:** II 56 available best (Fable 60 if available); expensive agent loops [C1][C7][C8].
- **Sonnet not default heavy:** II 53; ~$2.29/II task can exceed Opus because of verbosity [C8].

6. Escalation & routing rules

On a Grok-first factory, the first three steps share `cli=grok`:

```
light    grok  model=grok-composer-2.5-fast
  → standard  grok  model=grok-4.5 effort=medium
    → heavy   grok  model=grok-4.5 effort=high
      → heavy+  claude model=sonnet effort=high   # taste / Claude harness
        → frontier  claude model=opus effort=high   # +II ceiling; rare
```

- Escalate on **judgment** or **two verification failures** — not “prestige.”
- Composer → Grok 4.5 when the task needs deeper reasoning or multi-module judgment, not just speed.
- Do **not** auto-promote standard → heavy by switching to Sonnet; that often raises cost without a board-backed quality jump vs Grok high.
- Taste-sensitive work (skills, supervisor guidance, public docs, API naming) starts on Claude even if the diff is small.
- Prefer **model switch** over Claude effort above `high` (existing CAS rule; keep).
- De-escalate the tail: shut down Opus/Sonnet workers when only light tasks remain; sweep with Composer workers.

Task-shape cheat sheet

Task shape	Route
Mechanical edit, rename, small config, depth=light	**grok-composer-2.5-fast**
Bounded bug/feature, clear AC	grok-4.5 medium
Multi-module refactor, critical path	grok-4.5 high
Skill wording / public API / release prose	Sonnet → Opus
Architecture / ambiguous design	Opus (or Fable)
High-volume when Grok quota is tight	Codex low/medium backup
Failed twice on Grok 4.5	Sonnet then Opus

7. Composer 2.5 vs Haiku (corrected)

Composer is a first-class Grok Build model — use it for light

Verified on this host: `grok models` lists only `grok-4.5` and `grok-composer-2.5-fast` [C13]. Factory spawn is `cli=grok model=grok-composer-2.5-fast`. No Cursor harness required.

- **AA (Cursor CLI measure):** Composer 2.5 scored CAI 62 (3rd then), at ~$0.07 (standard) / $0.44 (Fast) per task, Fast wall ~6.7 min [C9].
- **Haiku:** still the fastest Claude decode (~97–127 tok/s) [C6], but loses on agent quality/$ vs Composer. Prefer Haiku only if you must stay on Claude for a tiny task.
- **Codex low:** still useful as bulk backup when Grok subscription/rate limits are the bottleneck — not as the preferred light model when Composer is available.
- **Note on AA harness:** AA measured Composer under Cursor CLI; factory will run it under Grok Build. Quality should be directionally similar; wall-time/$ may differ — treat AA as orientation, not identical factory telemetry.

8. What to ship in CAS (docs-first)

Routing is supervisor judgment today — no runtime model picker required for v1.

1. Update source of truth: `cas-cli/src/builtins/skills/cas-supervisor/references/model-selection.md`
2. Update hard-rule one-liner in `cas-supervisor/SKILL.md` (and Grok/Codex twins)
3. Mirror Grok twin under `cas-cli/src/builtins/grok/skills/cas-supervisor/`
4. Keep skill body under SessionStart 8KB budget; put numbers + model ids in the reference
5. Document model ids: `grok-4.5`, `grok-composer-2.5-fast` (from `grok models`)
6. Optional follow-on: change `WorkerSpec` stock defaults (separate product decision)

Suggested skill table (v2)

light = `grok / grok-composer-2.5-fast` · standard = `grok / grok-4.5 / medium` · heavy = `grok / grok-4.5 / high` · frontier = `claude / opus / high` · axes include Speed · cite AA snapshot date + `grok models` date.

Example spawns:

```
# light / flash workers
cas__coordination action=spawn_workers count=2 isolate=true \
  cli=grok model=grok-composer-2.5-fast

# standard workers
cas__coordination action=spawn_workers count=2 isolate=true \
  cli=grok model=grok-4.5 effort=medium

# heavy worker
cas__coordination action=spawn_workers count=1 isolate=true \
  cli=grok model=grok-4.5 effort=high worker_names="hv-ada"
```

9. Caveats

- **CAI scale drift:** May Composer scores (~62) vs July Grok prose (~76) may not be on identical index versions — trust relatives and $/tokens.
- **Harness effect:** AA measured Composer under Cursor CLI and Grok 4.5 under Grok Build. Factory Composer runs under Grok Build — expect directional, not identical, economics.
- **Only Fast listed here:** this host's `grok models` shows `grok-composer-2.5-fast`, not a separate standard Composer id. If xAI adds standard later, re-evaluate light pricing.
- **Subscription ≠ API $:** real factory burn often hits plan limits; token efficiency still matters either way.
- **Grok 4.5 TTFT ~16.5s:** weaker for single-shot chat; strong for multi-step agent loops [C2]. Composer Fast is the low-latency lane for small tasks.
- **Taste unbenchmarked:** keep Claude for human-facing writing.

10. Citations

[C1] Artificial Analysis, “Grok 4.5 brings SpaceXAI to the intelligence frontier” (8 July 2026). II 54; CAI 76 in Grok Build; ~$0.31/II task; ~$2.49–2.59/CAI task; ~1.9M tokens/CAI task; peer costs Fable 5 ~$11.80, GPT-5.5 Codex ~$5.07; ~14k output tokens/II task.
https://artificialanalysis.ai/articles/grok-4-5-brings-spacexai-to-the-the-intelligence-frontier

[C2] Artificial Analysis, Grok 4.5 (high) model page. II #4 / 54; output speed 91.3 tok/s; $2/$6; cache hit $0.50; TTFT 16.49s; 500k context.
https://artificialanalysis.ai/models/grok-4-5

[C3] Artificial Analysis models comparison / FAQ top ranks: Fable 5 (60), Opus 4.8 max (56), GPT-5.5 xhigh (55), Grok 4.5 high (54).
https://artificialanalysis.ai/models

[C4] Artificial Analysis, Claude Opus 4.8 (max) model page — list pricing $5/$25; speed band relative to peers.
https://artificialanalysis.ai/models/claude-opus-4-8

[C5] Artificial Analysis comparison: Claude Opus 4.8 (max) vs GPT-5.5 (medium) — II 56 vs 50; ~63 vs ~61 tok/s.
https://artificialanalysis.ai/models/comparisons/claude-opus-4-8-vs-gpt-5-5-medium

[C6] Artificial Analysis Anthropic provider page — Haiku ~97–127 tok/s; Sonnet 5 max ~84 tok/s ranking among Anthropic speeds.
https://artificialanalysis.ai/providers/anthropic

[C7] Artificial Analysis Intelligence Index v4.1 announcement notes — Opus max ~$1.78/II task; GPT-5.5 xhigh ~$0.99; decode time Opus ~6.4 min vs GPT xhigh ~3.7 min.
https://www.linkedin.com/pulse/announcing-artificial-analysis-intelligence-index-vzl8c

[C8] Artificial Analysis, “Claude Sonnet 5: strong agentic performance at a higher cost per task” (30 June 2026). II 53; ~$2.29/II task; ~15% more than Opus 4.8; ~40% more output tokens than Sonnet 4.6.
https://artificialanalysis.ai/articles/claude-sonnet-5-agentic-cost

[C9] Artificial Analysis, "Cursor's Composer 2.5: third on the Coding Agent Index…" (20 May 2026). CAI 62; $0.07 / $0.44 per task; wall time ~9.3 / ~6.7 min. Measured under Cursor CLI; quality/economics directional for Grok Build model id.
https://artificialanalysis.ai/articles/cursor-composer-2-5-coding-agent-index

[C10] Artificial Analysis Coding Agent Index methodology page (DeepSWE + Terminal-Bench v2 + SWE-Atlas-QnA; cost, tokens, wall time).
https://artificialanalysis.ai/agents/coding-agents

[C11] Anthropic, "Introducing Claude Opus 4.8" — list pricing $5/$25 standard; fast mode $10/$50.
https://www.anthropic.com/news/claude-opus-4-8

[C12] CAS in-repo: `cas-supervisor/references/model-selection.md` (current four-tier rubric); factory harnesses `claude|codex|grok` via cas-8888; `PtyConfig::grok` passes `-m` model through (`crates/cas-pty/src/pty.rs`); `WorkerSpec` grok-4.5 round-trip in `crates/cas-mux/src/spec.rs`.

[C13] Live host verification, 2026-07-09: `grok models` on Grok Build `0.2.93 (f00f96316d)` reports available models `grok-4.5` (default) and `grok-composer-2.5-fast` only. Confirms Composer is a Grok Build model id, not a separate harness requirement.